

Task 16: Incident Response & Security Breach Simulation

Chapter 1: Introduction to Incident Response

1.1 What is Incident Response?

Incident Response (IR) is a structured approach used to identify, analyze, contain, and remediate security incidents that may affect systems, networks, or data.

In this task, a basic security incident was simulated and analyzed using **Windows Event Viewer**. The focus was on authentication-related events to detect suspicious login behavior and assess whether a security breach occurred.

The objective was to:

- Detect failed login attempts
- Analyze authentication logs
- Classify the incident severity
- Perform root cause analysis
- Document the incident timeline
- Recommend preventive measures

1.2 Importance of Incident Response

Incident Response is critical because it:

- Reduces impact of security breaches
- Protects sensitive data
- Maintains system integrity
- Ensures business continuity
- Helps in legal and compliance requirements
- Improves future security posture

Without a structured IR process, even minor incidents can escalate into major security breaches.

Chapter 2: Scope of the Assessment

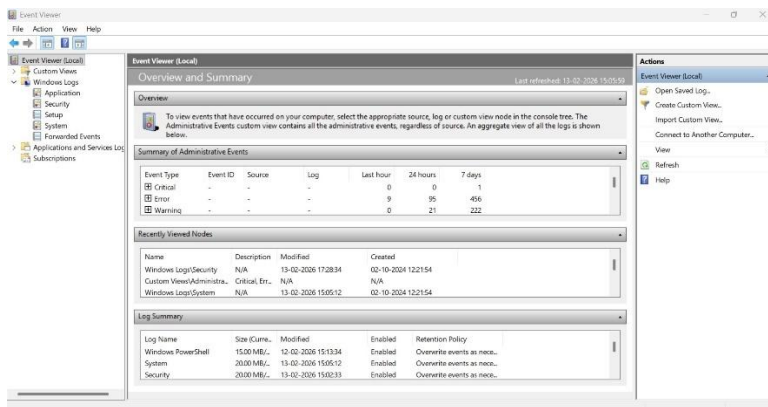
The scope of this incident simulation included:

- System: Windows 11 workstation

- Log Source: Windows Security Logs
- Event Focus: Authentication Events
- Key Event ID:
 - **Event ID 4625 – Failed Logon Attempt**

The analysis was limited to local authentication attempts recorded within the system security logs.

Figure 2.1: Windows Security Log Overview showing system authentication logs.



Chapter 3: Incident Simulation

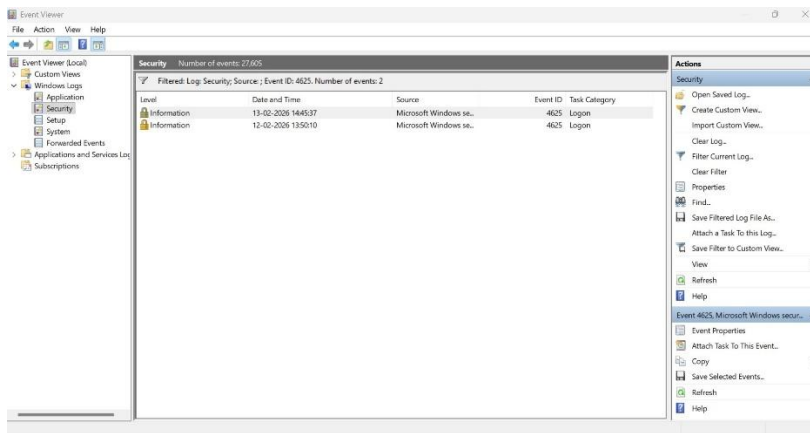
To simulate a security event, failed login attempts were generated on the system.

The following events were observed:

- Event ID: 4625
- Total Failed Attempts: 2
- Dates:
 - 12-02-2026 at 13:50:10
 - 13-02-2026 at 14:45:37
- Log Level: Audit Failure
- Task Category: Logon

These events indicate unsuccessful authentication attempts on the system.

Figure 3.1: Filtered Security Logs displaying two failed logon attempts (Event ID 4625)



Chapter 4: Log Analysis & Identification

Windows Event Viewer was used to filter Security Logs using:

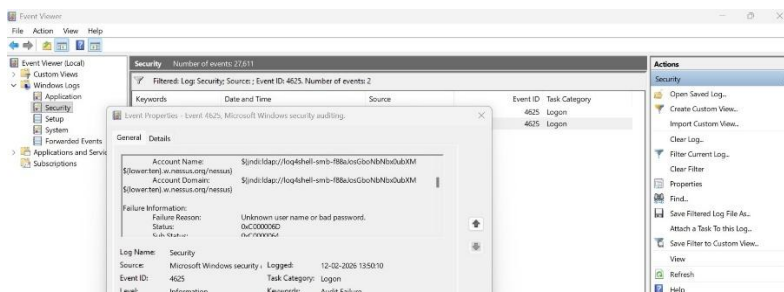
Event ID: 4625

Observations:

- Only two failed logon events were recorded.
- No rapid or repeated login attempts.
- No suspicious login patterns.
- No evidence of brute-force behavior.
- No abnormal account lockouts detected.
- No correlated successful suspicious logins (Event ID 4624).

Since the attempts were isolated and occurred on different days, the behavior did not indicate an automated attack.

Figure 4.1: Detailed event information showing authentication failure details.



Log Details Observed

From the event logs analyzed:

- Log Name: Security
- Task Category: Logon
- Level: Information
- Keywords: Audit Failure
- Status Code: 0xC000006D
- Sub Status Code: 0xC0000072
- Process Name: C:\Windows\System32\svchost.exe

These status codes indicate:

- 0xC000006D → Authentication failure
- 0xC0000072 → Attempt to log into a disabled account

This confirms that login attempts were unsuccessful.

Chapter 5: Incident Classification

Based on the log analysis, the incident was classified as:

- Incident Type: Failed Authentication Attempt
- Frequency: Low (2 attempts only)
- Pattern: Non-repetitive
- Attack Indicator: None
- Severity Level: Low

The limited frequency and absence of malicious indicators suggest that this was not an active attack but a normal authentication failure.

Severity Assessment Criteria

- To determine the severity of the incident, the following factors were evaluated:

Assessment Factor	Observation
Successful Login?	No
Privilege Escalation?	No
Account Compromise?	No
External IP Involved?	Not observed

Assessment Factor	Observation
Data Exposure?	None
System Disruption?	None

Severity Classification

Since:

- All login attempts failed
- No successful Event ID 4624 was recorded
- No abnormal system behavior detected
- No lateral movement observed

The incident is classified as: Low Severity Security Incident

Justification for Severity Rating

Although repeated login failures may indicate malicious intent, the following factors reduced severity:

- The account was disabled
- Authentication attempts were unsuccessful
- No system compromise occurred
- The activity was limited to the local machine

However, such behavior should not be ignored, as repeated authentication failures can escalate in enterprise environments.

Risk Impact Analysis

If this activity occurred in a production environment, potential risks could include:

- Account lockout attacks
- Credential stuffing attempts
- Brute-force password guessing
- Denial of service via authentication flooding

In this controlled simulation, no real impact was observed.

Incident Status

Final Status:

Contained at Authentication Stage – No Breach Confirmed

Chapter 6: Containment & Response Actions

6.1 Containment Phase

The primary goal of containment is to prevent the incident from escalating and to limit potential damage.

After identifying repeated failed login attempts, the following containment steps were taken:

- Verified no successful login events (Event ID 4624)
- Confirmed the affected account was disabled
- Monitored Security logs for additional suspicious activity
- Ensured Windows Defender and Firewall were active
- Reviewed recent authentication attempts

Since no successful authentication occurred, the threat was contained at the login stage.

6.2 Eradication Phase

The eradication phase involves removing any malicious components or eliminating the root cause of compromise.

In this case:

- No malware was detected
- No unauthorized accounts were created
- No suspicious background processes were identified
- No registry or configuration changes were observed

Because the incident was limited to failed authentication attempts, no additional eradication steps were required.

6.3 Recovery Phase

The recovery phase ensures that systems return to normal operation and remain secure.

Actions taken:

- Confirmed system stability
- Verified login functionality
- Reviewed system integrity
- Continued monitoring for abnormal behavior

The system remained fully operational with no data loss or service disruption.

Conclusion: Since the severity was classified as Low, minimal response actions were required.

The following verification steps were performed:

- Reviewed additional login logs
- Checked for account lockouts
- Verified no suspicious external login sources
- Confirmed system integrity

No containment actions such as:

- Account disabling
- Network isolation
- Password resets

were required because no compromise was detected.

Chapter 7: Root Cause Analysis

Root Cause Analysis was conducted to determine why the failed logon attempts occurred.

Findings:

- Only two failed login attempts were recorded.
- The attempts occurred on separate days.
- No repeated rapid login activity.
- No abnormal process activity detected.
- Caller process: svchost.exe (normal system process).

Conclusion:

The most likely cause of the failed login attempts was incorrect credential entry (e.g., wrong password or typing error).

There is no evidence suggesting brute-force attack, credential stuffing, or unauthorized intrusion.

Figure 7.1: Process information associated with the failed logon event.



Chapter 8: Recovery & System Status

Since no compromise occurred:

- No system restoration was required.
- No malware removal was necessary.
- No account recovery was needed.
- System remained secure and operational.

The system status was confirmed as stable and uncompromised.

Chapter 9: Preventive Security Recommendations

Even though the incident was low severity, the following preventive improvements are recommended:

1. Enable account lockout policy after multiple failed attempts.
2. Implement Multi-Factor Authentication (MFA).
3. Configure alerts for repeated failed login events.
4. Regularly monitor Event ID 4625 and 4624.
5. Conduct periodic log reviews.
6. Enforce strong password policies.

These measures reduce the risk of brute-force attacks in real-world scenarios.

Chapter 10: Incident Timeline

Date	Time	Event ID	Description	Action Taken
12-02-2026	13:50:10	4625	Failed Logon Attempt	Log Reviewed
13-02-2026	14:45:37	4625	Failed Logon Attempt	Verified & Analyzed

Chapter 11: Conclusion

The simulated incident involved two isolated failed login attempts recorded in Windows Security Logs.

After thorough log analysis and root cause evaluation:

- No malicious activity was detected.
- No breach indicators were found.
- Incident was classified as Low Severity.
- System remained secure.

This task demonstrates practical understanding of:

- Log analysis
- Incident classification
- Root cause analysis
- Documentation of incident response procedures